

THE ECONOMIC TIMES × ACCENTURE

ET AI Hackathon

Agentic Edition

Detailed Problem Statements

A guide for Hackathon Participants

HOSTED ON UNSTOP

ET AI Hackathon Problem Statements

Detailed guide for Hackathon Participants

Core structure of problems:

The core structure applies to all problem statements and is intended to help teams scope, develop, and present their solutions. It consists of three primary components:

1. Solution structure – 9 (3x3) blocker
2. Evaluation framework
3. Expectations from final submission

Solution structure:

All problems can be solved at multiple levels of complexity and maturity. To help teams define the scope of their solution, these levels are split across two dimensions/axes: solution depth on one axis and solution features on the other.

Solution depth will be a combination of two different aspects:

- a. The level of input uncertainty or lack of structure that the solution can handle
- b. The degree of reliability the generated output will be able to exhibit

Solution features will also be a combination of two different aspects:

- a. The number of relevant, meaningful/significant and distinct features in the solution
- b. The extent of supporting features embedded in the solution. (Those will typically be things like observability, trainability, fault tolerance, etc).

Each solution will therefore sit within a 3x3 grid that combines solution features and solution depth. Teams should use this grid to define the coverage they intend to demonstrate.

		Solution Features		
		F1	F2	F3
Solution Depth	D1			
	D2			
	D3			

For each individual problem statement, F1, F2, F3 and D1, D2, D3 are specified separately. Teams should review these requirements before deciding the scope of their solution.

Evaluation framework and approach:

The core approach towards evaluating the solutions would depend on the following key parameters:

1. **Significance and relevance:** Is the solution meaningful? Does it address a genuine business/user need connected to the core problem?
2. **Innovation & Originality:** How novel is the idea? Is there a genuinely creative approach rather than a conventional application of existing tools?
3. **Effective Use of AI:** Is AI central to the solution, or merely bolted on? Does the team exploit LLMs, agents, computer vision, ML, GenAI, etc. appropriately?
4. **Technical Complexity & Execution:** Quality of architecture, algorithms, AI pipelines, integrations, data handling, engineering, and implementation.
5. **Agentic / Autonomous Capability:** Where relevant, does the system reason, plan, use tools, take actions, recover from errors, and operate with appropriate autonomy?

6. **Business/User Impact:** Potential for measurable improvement: revenue, cost reduction, productivity, customer experience, risk reduction, sustainability, etc.
7. **Prototype Quality & Usability:** Does the prototype actually work? Is it intuitive and demonstrable?
8. **Scalability, Responsible AI & Robustness:** Can it scale? Does it address hallucination, security, privacy, bias, reliability, and human oversight appropriately?

The solution will also be mapped to the 9 blocker solution component grid to determine the extent of 'feature' and 'depth' covered. Teams should ensure that the working demo and detailed structural architecture provide clear evidence for the position claimed.

Solutions that can properly exhibit greater coverage will be rewarded accordingly.

Teams must declare their self-estimated position in the 9 blocker and clearly justify it through the final submission. There will be penalties for overestimation and underestimation.

Expectations from final submission:

The final submission must include two components. Together, they should clearly show what the solution does, how it works, and how the claimed solution features and solution depth are demonstrated:

1 – **A working demo.** It should illustrate the solution working across all areas where the team claims that it is working.

2 – **Detailed structural architecture:** Provide a detailed architectural breakdown of the solution showing the process flow, key actions, decisions, model usage, and incorporation of key features. The architecture should clearly explain how the solution supports the claims demonstrated in the working demo.

Problem 1 statement

Banking/financial regulations:

The banking industry is heavily influenced by regulations. Banks operate under a wide range of regulatory requirements that vary by geography, product and service lines, organizational structure, and other factors. These requirements also evolve over time.

Develop an agentic solution using one or more agents that understands regulatory requirements and how they change over time, assesses banks' internal control mechanisms and how they address those requirements, identifies gaps, and suggests appropriate remedies.

The core challenge is to maintain a continuously evolving and traceable relationship between:

Regulations → obligations → applicability → internal policies/controls → evidence → testing → gaps → remediation → ongoing monitoring.

Inputs to the solution:

1. Regulatory documents that are relevant to a particular bank
2. Documents related to internal controls, policies and related mechanisms

The following would be the core desirable features in the solution:

1. Regulatory intelligence and knowledge ingestion
2. Regulatory change intelligence
3. Regulatory obligation extraction
4. Bank control-framework understanding
5. Regulatory-to-control mapping
6. Control effectiveness assessment
7. Evidence-based compliance assessment

8. Gap identification
9. Risk-based gap prioritization
10. Remediation recommendation engine
11. Autonomous regulatory impact analysis
12. What-if / simulation capability
13. Cross-regulation intelligence
14. Regulatory contradiction detection

Solution depth requirements:

D1: Mostly structured and textual data as input. Should generate acceptable outputs in a majority of situations.

D2: Mostly structured and textual data as input. High degree of demonstrable reliability.

D3: Highly heterogeneous multimodal input. High degree of demonstrable reliability.

Solution features requirements:

F1: Covers at least 3 of the features from the feature list

F2: Covers at least 6 of the features from the feature list

F3: Covers at least 8 of the features from the feature list.

Problem 2 statement

Agentic Cybersecurity - Prompt Injection Firewall:

AI agents are used to take care of multiple standard activities in today's time. However, access to external content and tools also exposes them to malicious attempts to extract private information, perform unauthorized tasks, or trigger other unwanted behavior. One such technique is direct and indirect prompt injection.

A Prompt Injection Firewall is one of the most relevant cybersecurity problems today because many AI systems (LLM-based agents, copilots, and chatbots) can access external content and tools. The firewall's role is to

detect and neutralize malicious instructions embedded in user input or retrieved data before they influence the AI's behavior.

Design a *Prompt Injection Firewall* that intercepts all incoming content before it influences the AI's behavior. The solution should detect and neutralize malicious prompt injections while allowing legitimate content to pass with minimal disruption.

Input Sources

The firewall may receive:

- User messages
- Web pages
- PDFs
- Emails
- Markdown
- HTML
- Word documents
- API responses
- OCR text
- Source code
- Images (via OCR)

The solution should try to detect the following types of attack across the input sources listed above:

1. Instruction Override
2. Role Change
3. Secret Extraction
4. Tool Abuse
5. Credential Theft
6. Context Poisoning
7. Multi-Step Jailbreaks
8. Encoded Instructions
9. Indirect Prompt Injection

Solution depth requirements:

D1: Mostly structured and textual data as input. Should generate acceptable outputs in a majority of situations.

D2: Mostly structured and textual data as input. High degree of demonstrable reliability.

D3: Highly heterogeneous multimodal input. High degree of demonstrable reliability.

Solution feature requirements:

F1: Detect at least 2 types of attacks

F2: Detect at least 5 types of attacks

F3: Detect at least 7 types of attacks

Problem 3 statement

Retail - Autonomous Promotion Planner:

Retail promotions are a key tool for all retail companies to sell their products. However, successful promotion strategies require complex optimization, coordination, product and market understanding, and subjective judgments. Teams must account for these factors while balancing commercial objectives and operational constraints.

Develop an agentic planner capable of extracting, or inferring with a high degree of confidence, all relevant input parameters and making decisions on the key dimensions of the promo strategy. The planner should adjust to dynamic requirements as it builds the strategy and satisfy all necessary and relevant constraints imposed by the parent company.

Inputs to the solution:

The AI agent receives the following inputs:

- Inventory
- Competitor prices
- Holidays
- Margins
- Customer segments

Using these inputs, it decides:

- Which products to promote
- Discount amount
- Duration
- Target audience

These decisions must satisfy constraints such as:

- Minimum margin
- Inventory clearance
- Marketing budget

The following would be the core desirable features in the solution:

1. Select *which* products to promote
2. Determine the promotion mechanism
3. Optimize the discount and customize other strategies
4. Understand cannibalization
5. Understand product relationships
6. Consider inventory constraints
7. Geographically customize promotions
8. Competitor awareness
9. Simulate the promotion before launching

Solution depth requirements:

D1: Mostly structured and textual data as input. Should generate acceptable outputs in a majority of situations.

D2: Mostly structured and textual data as input. High degree of demonstrable reliability.

D3: Highly heterogeneous multimodal input. High degree of demonstrable reliability.

Solution feature requirements:

F1: Covers at least 2 of the features from the feature list

F2: Covers at least 5 of the features from the feature list

F3: Covers at least 7 of the features from the feature list.

Problem 4 statement

Utilities - Renewable energy orchestrator: Adoption and utilization of renewable energy are increasing and are expected to increase sharply in the future. However, renewable generation is highly variable. Coordination across solar, wind, batteries, and market prices is therefore critical to optimizing cost and sustainability.

Develop an agentic solution for renewable energy orchestration that coordinates solar generation, wind generation, battery storage, grid imports, energy prices, and demand forecasts. It should autonomously decide whether to charge the batteries, sell power to the grid, purchase from the market, curtail renewable generation, or take other appropriate actions as conditions change.

The final solution should be a **Renewable Energy Orchestrator:** an AI agent that autonomously manages a portfolio of renewable energy resources to minimize cost, maximize clean energy utilization, and maintain grid reliability.

Real-world scenario

Imagine a utility operating:

- 5 solar farms
- 3 wind farms
- 2 battery energy storage systems
- Several industrial consumers
- Grid interconnections
- Electricity market access

Every 15 minutes the situation changes.

Examples:

- Clouds reduce solar output
- Wind suddenly increases
- Electricity prices spike

- One battery becomes unavailable
- A transmission line reaches its capacity
- Industrial demand increases unexpectedly
- Weather forecasts are updated

The AI agent must continuously determine the best operating strategy as inputs, constraints, forecasts, and asset availability change. The working demo should show how the agent evaluates the situation, selects actions, and responds to changing conditions.

Inputs available to the agent

The agent has access to numerous data sources.

Renewable generation

- Current solar output
- Solar forecast
- Wind output
- Wind forecast

Grid

- Current demand
- Demand forecast
- Grid frequency
- Transmission constraints

Batteries

- State of charge
- Charging efficiency
- Maximum charge/discharge rate
- Battery degradation

Market

- Electricity prices
- Carbon prices
- Demand response incentives

External

- Weather forecast
- Storm alerts
- Maintenance schedules

Possible actions

Unlike a chatbot, the agent **takes actions**.

Examples include:

Battery actions

- Charge batteries
- Discharge batteries
- Reserve battery capacity for emergencies

Market actions

- Buy electricity
- Sell electricity
- Delay selling until prices rise

Renewable actions

- Curtail wind
- Curtail solar
- Prioritize cleaner generation

Demand management

- Trigger demand response
- Reduce noncritical loads
- Shift industrial loads

Maintenance

- Delay maintenance
- Schedule maintenance
- Dispatch inspection teams

Objectives

Unlike classical optimization, there are **multiple competing goals**.

For example:

Minimize

- Electricity cost

- Carbon emissions
- Renewable curtailment
- Battery degradation

Maximize

- Reliability
- Renewable utilization
- Profit
- Grid stability

These multiple competing goals naturally create trade-offs. The solution should make the selected optimality criteria clear and show how actions are chosen when objectives conflict.

Solution depth requirements:

D1: Mostly structured and textual data as input. Should generate acceptable outputs in a majority of situations.

D2: Mostly structured and textual data as input. High degree of demonstrable reliability.

D3: Highly heterogeneous multimodal input. High degree of demonstrable reliability.

Solution feature requirements:

F1: Determine optimal cluster of actions

F2: Determine optimal cluster of options after determining optimality criteria given a specific situation and adjusting for uncertainty

F3: Simulate situation/scenario over a period of time by accounting for potential variations in the input conditions.